Generation and Measurement of a Reference Field for Round-Robin Comparison Purposes

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***Abstract* – An accurately known reference field is often the reference quantity in round-robin comparisons of electromagnetic field measurements. The limits of accuracy inherent to the definition and measurement of a reference electromagnetic field are here discussed and quantified in the case of measurements at 3 m distance from the field source and in the 30-300 MHz frequency range. The insertion loss between the transmitting and receiving antenna is considered here as an alternative reference quantity. A method of validation of insertion loss predictions delivering an accuracy of a few tens of dB is described and experimentally demonstrated. Reference to a practical selection of field source and receiving antenna is made for general applicability of the results here obtained.**

***Keywords* – Round-robin comparison, electromagnetic field measurements, reference electromagnetic field, measurement uncertainty.**

I. INTRODUCTION

A round-robin comparison of electromagnetic field measurements in compact fully anechoic rooms (CFARs) started in 2005 and involved 14 sites up to 2007 [1]. CFARs are alternative test sites designed and operated for electromagnetic field measurements, immunity and emission tests, at 3 m distance from the source of radiation. The scope of the round-robin comparison is to estimate the deviation of the measurements from site to site due to the CFARs non idealities, the ideal being empty space. To eliminate spurious contributions from the instrumentation to the site-to-site variations, all the measuring equipment is itinerant, so that the stability of the traveling equipment becomes the only limit to the resolving power. The frequency range investigated is 30-300 MHz.

Although the scope of the round-robin comparison is not to compare the field measured with the one predicted in an ideal site, a knowledge of the field impinging on the receiving antenna is of great advantage for the design of the round-robin itself, in particular for the selection of the receiver's settings relevant to sensitivity (noise level) and dynamic range. Also, the availability of an accurately known reference field allows the correlation of the geometrical and physical properties of the CFARs with their performances, if a relatively large sample of participants is available.

The aim of this paper is to describe how a reference electromagnetic field suitable for round-robin comparison purposes in the 30-300 MHz frequency range can be generated and accurately predicted using standard laboratory instrumentation. Sections II and III are devoted to the description of the field source and the receiving antenna, respectively, focusing on the analysis of their relevant radiation properties. In section IV the selection of the more appropriate reference quantity is discussed, mainly whether the field radiated by the source or the insertion loss between the field source and the receiving antenna is to be preferred. In section V the experimental evaluation of the accuracy of the predictions is presented and the results are discussed. Conclusions follow.

II. THE FIELD SOURCE

The field source consists of a self-constructed comb generator feeding a 30 cm length, 4 mm diameter rod (the monopole). The metallic box housing the comb generator acts as the counterpoise of the monopole, thus the whole radiating structure is equivalent to an asymmetric dipole. The box dimensions are L=12 cm x W=6.8 cm x D=3 cm, the L dimension being parallel to the monopole axis.

The electronic circuit of the comb generator is described in [2] and it is based on the use of a Step Recovery Diode for forming and shaping a very short pulse at a relatively high repetition rate. Our realization conforms to that in [2] except for the adoption of a permanent PCB realization (instead of a perfboard with ground plane) and for the use of recent technology, off-the-shelf, switching logic circuits. The frequency spacing between the harmonics is 5 MHz thus providing 55 spectral lines from 30 to 300 MHz. The comb generator power P_0 delivered to a 50 Ω load is relatively flat in this frequency range, being −33 dBm ± 1 dB (see Fig. 1), and greater than −40 dBm up to 700 MHz.

The output impedance Z_S of the comb generator (nominally 50 Ω) was measured with a Vector Network Analyzer. Due to an internal pad, of about 8 dB, at the output of the comb generator, the output impedance Z_S is relatively flat and smooth with frequency, and is essentially resistive, de-

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creasing monotonically from about 65 Ω to 40 Ω, 30 to 300 MHz.

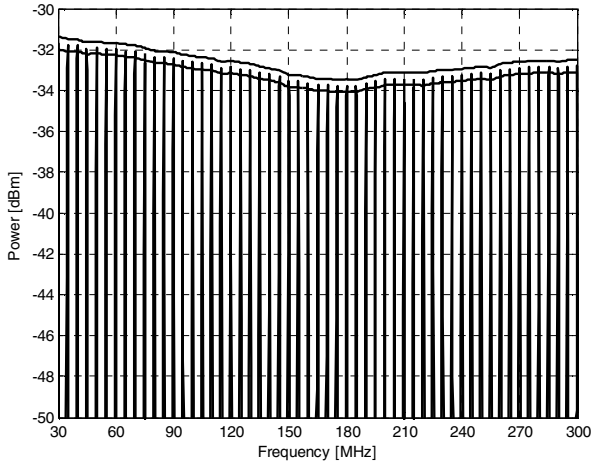


Figure 1. Comb generator power delivered to a 50 Ω load (±0.3 dB accuracy, with 95% confidence level).

The stability of the generator is 0.05 dB (1 standard deviation), including intrinsic fluctuations and the effect of the thermal variations within limits reasonably expected in indoor sites (±5 °C).

The radiating structure was analyzed with a commercial electromagnetic simulator based on the Method of Moments (MoM). For numerical computation purposes the monopole was subdivided into discrete cylindrical segments while the surface of the box into triangular patches. The dimension d of the discrete elements - height for segments, and side for patches (approximately equilateral triangles) - is short with respect to the minimum wavelength of interest so that assuming uniform current along them is legitimate. In particular we set $d = \lambda_{min}/40$, where λ_{min} is the minimum wavelength, i.e. λ_{min} = 1 m (300 MHz maximum frequency), then d = 2.5 cm.

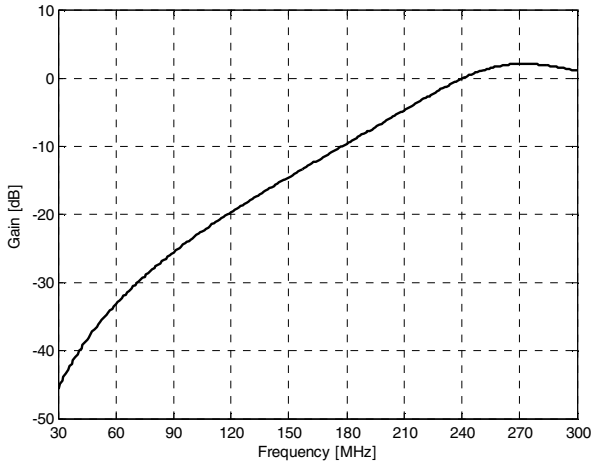


Figure 2. Numerically computed realized gain of the radiating source.

The computed realized gain G of the radiating structure is plotted in Fig. 2 as a function of frequency. The radiating structure behaves as a short dipole in the low-frequency range (gain increasing at 40 dB/dec, below resonance) and it is resonant at about 270 MHz where it achieves its maximum realized gain value (nearly +2 dB).

The field radiated perpendicularly to the monopole axis at 3 m distance from the point where the monopole axis crosses the metallic box, can be obtained as

$$E' = G + P_0 - 20\log\left(\left|1 - \Gamma_S \Gamma_R\right|\right) + 95.23 \qquad (1)$$

where E' is the electric field in dB(μV/m), P_0 is the power delivered by the comb generator to a 50 Ω load (in dBm), Γ_S and Γ_R are the reflection coefficients looking into the generator and the antenna terminals, respectively. Equation (1) is valid in the far field of the source, a condition not rigorously met in the lower frequency range at 3 m distance. The deviation $E' - E$ between the field predicted by (1) and the field E, numerically computed by using MoM is presented in Fig. 3. Note that here the reactive components of the field are taken into account. The deviation decreases with increasing frequency from about 1 dB at 30 MHz to 0.05 dB at 300 MHz. The prediction of the field as given by (1) relies both on numerical computation (for the determination of G and Γ_R) and on measurements (for the determination of Γ_S).

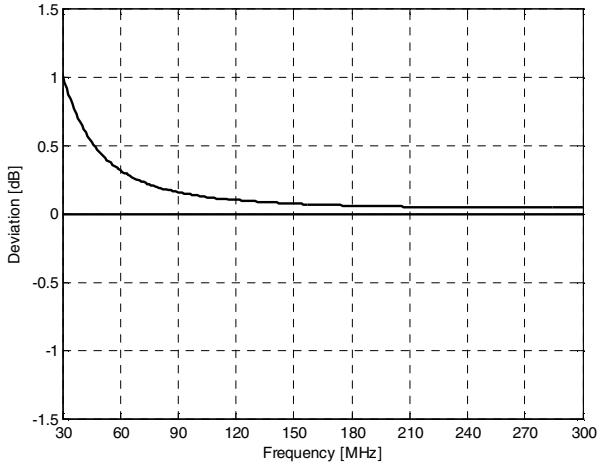


Figure 3: Deviation between the electric field E' calculated through equation (1) and the numerically computed electric field component E parallel to the field source axis at 3 m distance.

III. THE RECEIVING ANTENNA

The receiving antenna is a Schwarzbeck biconical antenna with radiating elements model BBA 9106, and 1:1 balun model VHA 9103. The length of the antenna is 1.31 m and the diameter of its wired cones is 52 cm. The inversion test of the antenna exhibited a maximum deviation of 0.4 dB, and 0.14 dB root-mean-square (rms) over the 30-300 MHz frequency range.

The radiating structure was analyzed by means of MoM to numerically predict its free-space antenna calibration factor (*ACF*). The dimension d of the discrete segments and patches

composing the simulated structure is $d = \lambda_{min}/40$. The short-circuit between the axial rod and one of the six wires forming each cone is included in the simulated structure. The removal of the resonance that would be otherwise present at 287 MHz [3] if the short-circuit was absent is correctly predicted. The metallic box holding the balun circuit is also included to take into account possible coupling effects with the field source in experiments where they face each other in close proximity (see section V).

The free-space antenna factor cannot be correctly determined solely relying upon the numerical simulation of the radiating structure. Indeed the balun circuit includes a multi-turn coil, wound around a ferrite core and the value of the impedance seen by the cones when looking into the balanced terminals of the balun is significantly affected by the loading of this coil. In the following we will refer to this loading effect as the "balun loading". The impedance seen by the cones is plotted in Fig. 4. Note that the deviation from the unloaded 50 Ω value is significant.

The balun action is obtained as follows (a scheme of the Schwarzbeck balun is in Fig. 3 of [4]). Two micro-coaxial cable sections (RG 178 U type) are symmetrically wound across each hole of a double aperture ferrite core (see the inset in Fig. 6) and the inner conductor of one of the cables is connected to the braid of the other one in correspondence of the gap between the two. The braids of the micro-coaxial cables are short circuited at the opposite side of the gap, with respect to the core. The radiating elements are connected to the braids of the two cables at the gap. The balun action is the same as that provided by the shield of a shielded loop probe [5], the only differences being: a) the presence of the ferrite core, around which the micro-coaxial cables are wound; b) the radiating structure is the loop shield in the case of the probe, while it is the cone pair in the case of the antenna being considered here.

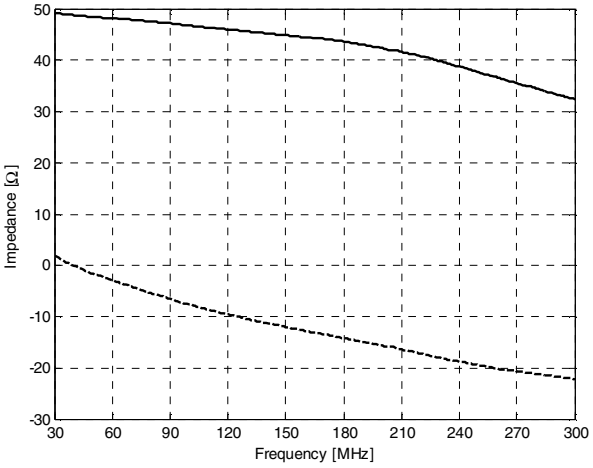


Figure 4. Impedance seen by the cones of the receiving antenna when looking into the balanced terminals of the balun. Continuous line: resistance, dashed line: reactance.

The impedance Z_B seen by the radiating elements looking into the balanced terminals is thus given by

$$Z_B = Z \,||\, 50\,\Omega \qquad (2)$$

where Z is the coil impedance, and 50 Ω represents the input impedance of the receiver connected to the unbalanced terminals. The presence of the ferrite core is needed to raise the impedance Z to a sufficiently high value with respect to 50 Ω, to prevent excessive reduction of the sensitivity of the receiving antenna.

The impedance Z, which is responsible for the balun loading effect, was determined according to the simplified model in Fig. 5. Impedance measurements were performed at the unbalanced port of the balun (left side port in Fig. 5) with the balanced terminals (right side port) left open (the cones are not connected during impedance measurements). The effect of the cables connecting the unbalanced port to the terminals of the coil was extracted from impedance measurements. The cables are RG 58C/U and RG 178/U, the micro-coaxial realizing one of the two windings and their lengths are $l_{58} = 50$ cm and $l_{178} = 8.18$ cm. Fig. 6 shows the plot of the amplitude of the impedance Z for frequencies between 500 kHz to 100 MHz (dots). It is interesting to observe that the coil behaves as an inductance for frequencies up to 10 MHz (inductance value 2.4 μH, corresponding to the continuous line in Fig. 6, to be compared with 2.1 μH ± 25% as derived from the data sheet of the core) and it resonates at about 30 MHz, i.e. the start of the frequency range of operation of the balun.

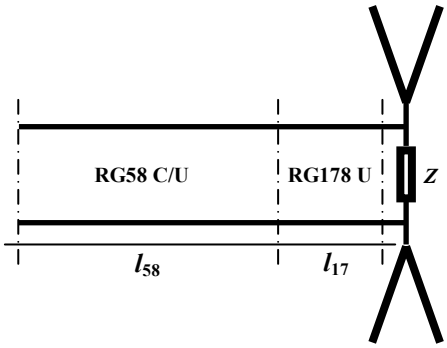


Figure 5. Simplified model used for the determination of the coil impedance Z from impedance measurements at the unbalanced terminals of the balun (left-side port). The impedance Z is connected to the balanced terminals (right side port), in parallel with the cones of the biconical antenna.

The ACF of the receiving antenna is represented in Fig. 7 and it was obtained through the following two steps:

a. The radiating structure is analyzed by means of MoM, assuming that the cones see a 50 Ω load looking into the balanced terminal of the balun. The antenna calibration factor ACF_0, and the input impedance Z_C of the conical dipole are thus numerically obtained;
b. ACF is derived through the following formula

$$ACF = ACF_0 + 20\log\left(\left|1+\frac{Z_C}{Z_B}\right| \Big/ \left|1+\frac{Z_C}{50\,\Omega}\right|\right) + A \qquad (3)$$

where the second term at the right side accounts for the balun loading effect, and A is the total attenuation of the two sections of cable inside the balun. Note that, similarly to the derivation of (1), the value of ACF as given by (3) is in part obtained through numerical computation (ACF_0 and Z_C) and in part through measurements (Z_B).

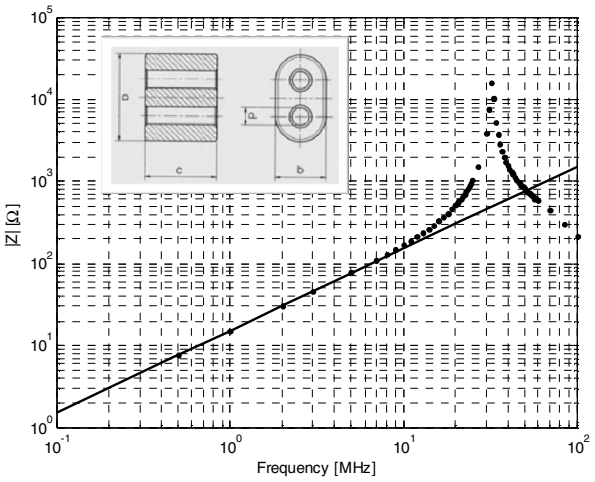


Figure 6. Impedance Z of the coil of the balun in the 500 kHz to 100 MHz frequency range (dots). The continuous line increases at 20 dB/dec and is obtained through a least-square fit of the coil inductance from the first five measured impedance values. The inset represents the double aperture ferrite core used in the Schwarzbeck balun. The core dimensions are a = 14.4 mm, b = 8.5 mm, c = 14.5 mm, d = 3.5 mm.

The frequency behavior of ACF (continuous line) and of ACF_0 (dashed line) are plotted in Fig. 7 where the manufacturer calibration data (dots) are also shown. It is interesting to observe that the balun loading is negligible at low-frequencies and becomes significant above 150 MHz, where the deviation between ACF and ACF_0 reaches about 1.5 dB at 300 MHz. Note also that, at high frequencies, the calibration data (typical, stated with ±0.7 dB accuracy, with 95% confidence level) better fit ACF_0 rather than ACF. Such discrepancy goes apparently against common sense and its origin is not evident.

Several concurrent non-ideal aspects might be invoked to justify the unexpected coincidence appearing in Fig. 7. The tolerance of some balun components is in certain aspects rather large, e. g. the magnetic properties suffer from a range of variation of ±25%. The manufacturer's calibration procedure (height scanning and averaging), although non-standard, is however to be considered highly credible in that it was authoritatively confirmed by the round robin comparison in [6] according which our antenna manufacturer proved to be capable of providing the free-space antenna factor of VHF antennas with ±0.5 dB accuracy (95% confidence level). Our specific antenna might have fallen somewhat out of the central portion of the calibration data distribution on which the specifications are based. In fact, interpretation of the available calibration data [7] suggests that an uncertainty somewhat above 1 dB can be envisaged. Admittedly, we are here paradoxically trying to justify agreement rather than disagreement on the basis of uncertainties or tolerances.

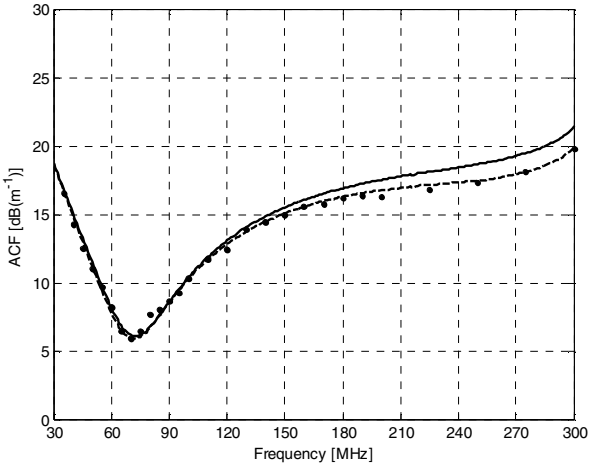


Figure 7. Antenna calibration factor (ACF) of the receiving antenna (continuous line), neglecting balun loading (dashed line), manufacturer calibration data (dots).

In the experimental section V it is shown that neglecting balun loading leads to an increased deviation between prediction and measurements. It is thus confirmed that this ingredient should not be missing in the recipe for an accurate ACF prediction.

IV. INSERTION LOSS VS. ELECTRIC FIELD

The electric field radiated by the field source is typically the reference quantity in round-robin comparisons. The reason is that typically the goal of round-robin comparisons is to evaluate the reproducibility of field measurements. Therefore each participant measures the field radiated by the source with its own antenna (as well as the receiving instrument, the cables, etc.). In cases such as in this paper where the antenna is not changed, the insertion loss between the field source terminals and the receiving antenna terminals is the most convenient reference quantity. In fact, the electric field radiated by the field source at 3 m distance in the 30-300 MHz frequency range cannot be quantified with high accuracy because:

a. Non negligible reactive components are present superimposed to the radiated one (given by (1)), in particular in the low-frequency range (see section II, last paragraph);
b. Non negligible mutual coupling exists between the field source and the receiving antenna, especially at high frequencies;
c. The field along the axial dimension of the receiving antenna is not uniform due to the non-isotropic, frequency dependent, pattern of the field source, and to the amplitude and phase variations of the component of the electric field along the axis of the receiving antenna.

The cumulative effect of these unwanted sources of inaccuracy can be quantified through numerical computation, by comparing the MoM predicted electric field E with the field E'' obtained from the following equation

$$E'' = V_M + ACF = P_0 - IL + 107 + ACF \qquad (4)$$

where V_M is the voltage that would be measured by a 50 Ω receiver connected to the receiving antenna terminals and *IL* is the numerically computed insertion loss, which takes into account all the above listed effects a, b and c. According to the definition of antenna calibration factor the field E'' is the one expected to impinge on the receiving antenna. In Fig. 8 it is shown the difference $E'' - E$ in the case where the field source and the receiving antenna are those described in sections II and III. E is calculated at the point corresponding to the phase center of the receiving antenna and it is the field that would be measured by an elementary electric dipole.

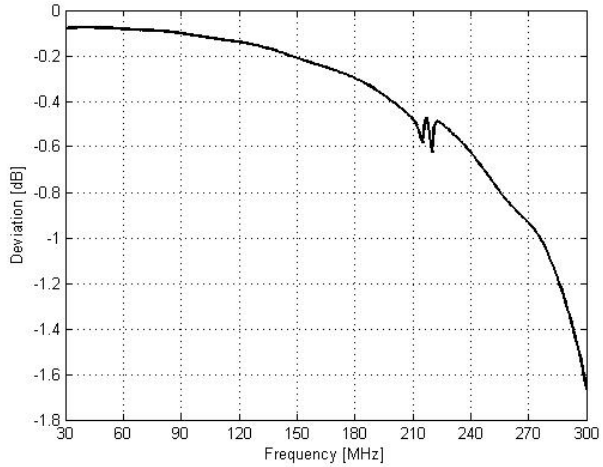


Figure 8. Deviation between the electric field *E* predicted by means of MoM and that derived using (4).

Note that $E'' < E$ and that the deviation reaches −1.6 dB at 300 MHz, a relatively large value. We conclude that *IL* does not suffer from ambiguities in its definition, as it would be the case of the electric field radiated by the source, and therefore it is the suitable quantity to be used as reference at least in the case where the emitter and receiver are itinerant among participating laboratories.

V. EXPERIMENTAL CONFIRMATIONS

Several experiments were performed to prove the validity of our predictions. We here present the results of insertion loss measurements between a pair of biconical antennas and between the field source and a receiving biconical antenna. We compare experimental results with predictions, the deviation between the two being a measure of the reliability of predictions.

The experiments are performed in an open-area site for pre-compliance measurements. The site lies in a natural environment, the main reflecting objects outside the obstacle-free area (a rectangle of about 10 m length x 5 m width) are trees spaced several meters apart and sparse vegetation. The ground is clay soil. To minimize the effects of multiple reflections and scattering from the nearby reflecting objects, and of the coupling of antennas with ground, the distance between the emitter (biconical antenna or field source) and the receiving antenna is set to a minimum practical value, i.e. less than 1 m in both cases. The polarization of the transmitting and receiving antennas is vertical. Because of the strong coupling between the antennas particular care is provided in order to assure their accurate placement and alignment. Also, special care is taken for the emitter/receiver positioning (especially the wired cones), the same one that is adopted in the numerical computations. The accuracy of the insertion loss predictions evaluated through these very short distance experiments is deemed to represent a challenging situation, that is, much more physically demanding in comparison to ordinary transmission/reception measurements at 3 m distance.

A. Insertion Loss between the Biconical Antennas

The comparison between measurement and prediction is reported in Fig. 9. The two biconical antennas are nominally identical, being of the same type as that reported in section II. The distance between the antennas is 93.7 cm, their height above ground is 1.59 m, the antenna phase center being the reference point. The cables are run horizontally for about 2 m behind the antennas before letting them fall vertically to ground.

Free space is assumed in the predictions. Also the case where a perfect conducting ground is present was considered in computations but its effect is negligible in comparison to the strong direct coupling between the antennas.

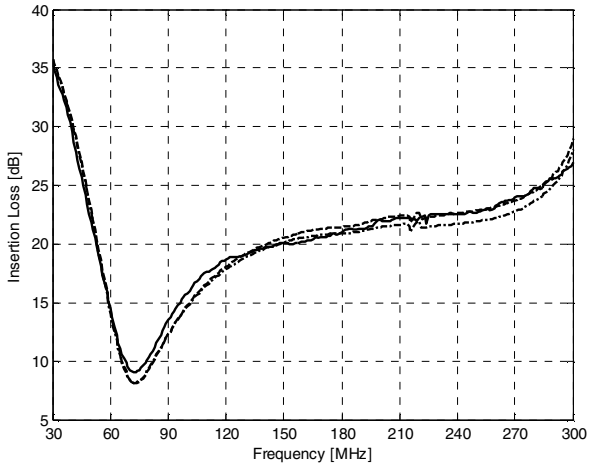


Figure 9. Insertion loss between two nominally identical biconical antennas. Continuous line: measurement, dashed line: prediction, dash-dot line: prediction with balun loading neglected.

The maximum deviation in the 30-290 MHz frequency range is 1.2 dB, the rms deviation is 0.6 dB. From 290 to 300 MHz the deviation increases reaching 3.2 dB. In this last 10 MHz frequency range a steeper increase of insertion loss is predicted than the one experimentally observed. This increase is due to the cone resonance being shifted from 287 MHz to 305 MHz by the short-circuit bar. The effect of this resonance is more pronounced when the antennas are very close together. The predicted damping of the resonance at 305 MHz is lower than the one measured because conductor losses are neglected

in the numerical computations. This phenomenon tends to become less important when the distance increased to values greater than 1 m. Thus we conclude that if a hypothetical experiment was performed in free space, with the antennas placed at 3 m distance, the agreement would be better than 0.6 dB rms through the whole 30-300 MHz frequency range. The dash-dot line in Fig. 9 corresponds to the insertion loss predicted neglecting balun loading. Note that an appreciable systematic deviation of about 1 dB between prediction and measurement appears above 200 MHz when neglecting loading thus demonstrating that balun loading provides a not negligible contribution to improving prediction accuracy.

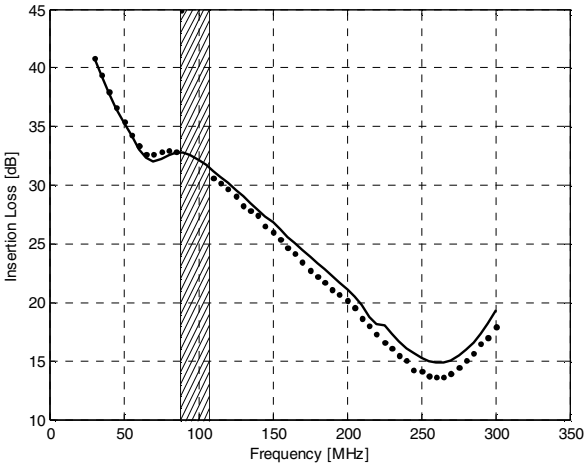


Figure 10. Insertion loss between the field source and the biconical antenna when placed at 30 cm distance. Dots correspond to measurement, continuous line to prediction. Data missing in the shadowed area because of strong disturbing radiation (FM broadcasting).

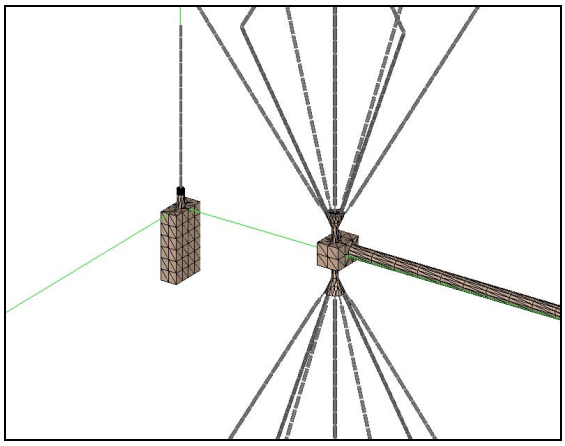


Figure 11. Picture of the field source and the biconical antenna generated by the electromagnetic simulation program corresponding to the experiment described in sub-section V.B. Segments and patches of the radiating structures are evident.

B. Insertion Loss between the Field Source and the Biconical Antenna

The insertion loss between the field source and the biconical antenna (one of the two in the previous experiments, subsection V.A) is shown in Fig. 10. The field source and the biconical antenna are placed at 30 cm distance, and their height above ground is 2.10 m. The shadowed area corresponds to FM broadcasting radiation. Numerical computations are again performed assuming free space. A graphical representation of the situation is in Fig. 11 as generated by the simulation program. The maximum deviation between prediction and measurement is 1.3 dB, the rms deviation is 0.8 dB. The agreement below 150 MHz is excellent (0.7 dB maximum deviation). An appreciable systematic deviation appears above 110 MHz ranging from 0.4 dB to 1.3 dB, which is however a satisfying result if one appreciates that it is obtained in an extremely challenging configuration and in a pre-compliance site using ordinary instrumentation.

VI. CONCLUSIONS

It is here demonstrated that accuracies of the order of a few tens of a dB can be achieved in the prediction of the insertion loss between two antennas by using a commercial electromagnetic simulation software, a pre-compliance test site and ordinary measurement instrumentation. Insertion loss is an unambiguous quantity which can be measured with great accuracy through comparison with calibrated attenuators. Due to these characteristics insertion loss is particularly suitable to be taken as a reference for round-robin comparison purposes, at least when the emitter and the receiver are itinerant among participants. Measurements performed with the emitter and the receiver at very short distance, so as to reduce the unwanted effects originating from site imperfections, can be used to validate predictions of insertion loss.

ACKNOWLEDGMENT

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